THE PROBLEM

AI agents can pay. They can’t manage API keys.

Human-shaped billing breaks agents

- Subscriptions and dashboards assume a human operator
- API keys are secrets agents shouldn’t hold long-term
- Per-call on-chain gas kills micropayments

What machine-to-machine needs

- Escrow once, pay incrementally
- Cryptographic vouchers, not passwords
- Provider batch settlement without trusting the agent after deposit

WHY PORTALDOT V3

EVM apps on a Substrate-native stack

Execution

Portaldot V3 exposes an EVM-compatible layer (pallet-revive). Solidity contracts ship with familiar tooling — Foundry, cast, viem — while staying on Portaldot's testnet.

Native tPOT tabs

AgentTab escrows native **tPOTv3** (14 decimals on Substrate) or any ERC-20. Middleware for agent payments fits the Modular Cloud track.

Interop path

Same AgentTab bytecode from a native `5…` account via `ReviveApi_call` / `Revive.call`. Probe matches `eth_call`. A submitted extrinsic is still TODO.

Network facts

Chain ID `420420777`
RPC `test-evm-rpc-v3.portaldot.dev`
Console node-console.portaldot.dev

SOLUTION

AgentTab — escrow + EIP-712 vouchers

Agents escrow tPOT once, sign a voucher per API call; providers batch-settle on Portaldot V3 — micropayments without API keys or per-call gas.

open

Payer deposits native tPOT (or ERC-20) for a provider, with expiry. One tab, one escrow.

voucher

Each call: agent signs EIP-712 `(tabId, cumulative)`. Provider returns HTTP 402 if missing/invalid.

settle

Provider submits cumulative + sig once for N calls. Contract pays `cumulative - claimed`.

refund

After expiry, payer reclaims unspent balance. Reentrancy guard + existence checks on every path.

ARCHITECTURE

Agent → Provider → AgentTab on Portaldot V3

01
AI agent
open(provider, NATIVE, deposit, expiry) with tPOT value

→

02
API provider
request + voucher sig; respond or HTTP 402

→

03
AgentTab.sol
settle batches N calls; refund after expiry

Status for the 2026-11-06 freeze

Contract live (`forge test` recorded 7/7). Gateway + SDK settled tab #2 on 2026-10-23. Read-only dashboard in `dashboard/`. Substrate `ReviveApi_call` matches `eth_call` for `remaining(1)`. Judge path: `STATUS.md` · `docs/FINAL-DEMO.md`.

LIVE TESTNET PROOF

Deployed on Portaldot V3 · chain 420420777

Contract	`0x9a096C216f79e08C2DB95d92F412697D8F1f9329`
Deploy tx	`0xe8a4376602a5467574dd7f2e966006da4fe41275d28a989e33f116fa57baab72`
Demo open	`0xc27ab271b49103a257046c7458805ed1c73988d464f61aa98bfe6338a13e534e`
Demo settle	`0xd58678d04863861cda814284eb2879d0abcaf85ca64779cc5e69f2a670b8ac5b`
Deployer	`0x9e2e2024c7dcA86e57608e2ceAD160081c381fDd`
Explorer	node-console.portaldot.dev · see also `DEPLOYMENT.md`

Native tPOT tab #1: deposit 1e16 wei → settle cumulative 5e15 wei via EIP-712 → remaining 5e15.

DEMO FLOW

open → voucher → settle (on-chain today)

OPEN
Tab #1
payer = deployer
provider = 0x7F22…3369
deposit = 1e16 wei native

→

VOUCHER
EIP-712
agent signs (tabId=1, cumulative=5e15)
off-chain, zero gas

→

SETTLE
Provider tx
pays 5e15 to provider
remaining(1) == 5e15
hash 0xd58678d0…ac5b

Reproduce locally

```
forge test && cp .env.example .env && scripts/deploy.sh
```

Gateway + SDK, 2026-10-23: tab #2 open `0x9922d9a7…f886`, settle `0x2031c40d…df99` (0.0001 tPOT). Hashes: `DEPLOYMENT.md` · script: `docs/FINAL-DEMO.md` · Week 2 replay: `docs/demo-checkin.mp4`

SHIPPED · FREEZE 2026-11-06

What a judge can check

Gateway + SDK settle

HTTP 402 quote, EIP-712 voucher, on-chain settle of tab #2. `npm test` and `npm run dry-run` need no key.

Dashboard

Static inspector. `npm run dashboard` reads tabs from the public RPC. A public dashboard URL is still TODO.

Substrate interop

`npm run revive-probe`: same `remaining(1)` via ReviveApi and eth_call. Assets precompile address for asset 2. Unsigned `Revive.call` only — extrinsic TODO.

Links still TODO

Published GitHub Pages URL, and a hosted final video. This deck + `docs/demo-checkin.mp4` are the Week 2 assets (title cards, not a gateway recording).

ASK

Continued support & office hours

Where to look before 2026-11-06

1) `STATUS.md` and `docs/FINAL-DEMO.md` — shipped surfaces and the five-minute path
2) Live gateway settle: tab #2 `0x2031c40d…df99` on chain 420420777
3) This deck, plus Week 2 video `docs/demo-checkin.mp4` (a hosted final cut is still TODO)

Public repo · live contract · demo txs already on-chain.
github.com/zwang1111/agenttab-portaldot · BUIDL 50150

Thank you — AgentTab × Portaldot